**D 51772** **(Pages : 2)** **Name.....**

**Reg. No.....**

**THIRD SEMESTER (CBCSS—UG) DEGREE EXAMINATION**
**NOVEMBER 2023**

Physics/Applied Physics

PHY 3C 03—MECHANICS, RELATIVITY, WAVES AND OSCILLATIONS

(2019—2022 Admissions)

Time : Two Hours Maximum : 60 Marks

*The symbols used in this question paper have their usual meanings.*

**Section A (Short Answer Type)**

*Answer* **all** *questions in two* **or** *three sentences.*
*Each correct answer carries a maximum of* 2 *marks.*

1. Give the significance of mass energy relation.
2. Define with example, fictitious forces.
3. What is a conservative force ?
4. Explain length contraction.
5. What is the purpose of two stages in a rocket ?
6. Give one application of Newton's third law.
7. What is the relation connecting length and period of oscillation of a simple pendulum.
8. Give three examples for periodic motion.
9. Define a black body.
10. What are matter waves ?
11. What are the factors on which Photo electric current depends ?
12. Explain the term threshold potential.

(Ceiling - 20)

**Turn over**

**Section B (Paragraph / Problem Type)**

*Answer* **all** *questions in* **a paragraph** *of about* **half a page to one page**.
*Each correct answer carries a maximum of* 5 *marks.*

13. What is a conservative force ? Show that a central force is always conserved.
14. Show that a conservative force is represented by the relation F = -grad U, and curl of it is zero.
15. Find the momentum of an electron which is accelerated by a potential difference 20 volt.
16. Explain the principle of rocket propulsion.
17. With figure explain the scattering of a particle by a heavy nucleus.
18. Write the equation and explain the variation of mass with velocity.
19. Explain the terms Eigen values and Eigen vectors.

(Ceiling - 30)

**Section C (Essay Type)**

*Answer any* **one** *in about* **two pages.**
*Answer carries* 10 *marks.*

20. Define Damped harmonic oscillator. Derive the general equation.
21. Write law of conservation of angular momentum. Illustrate with :
    a) Satellite motion ;
    b) Scattering of $\alpha$ particle by a heavy nucleus ; and
    c) Shape of galaxy.

(1 × 10 = 10 marks)